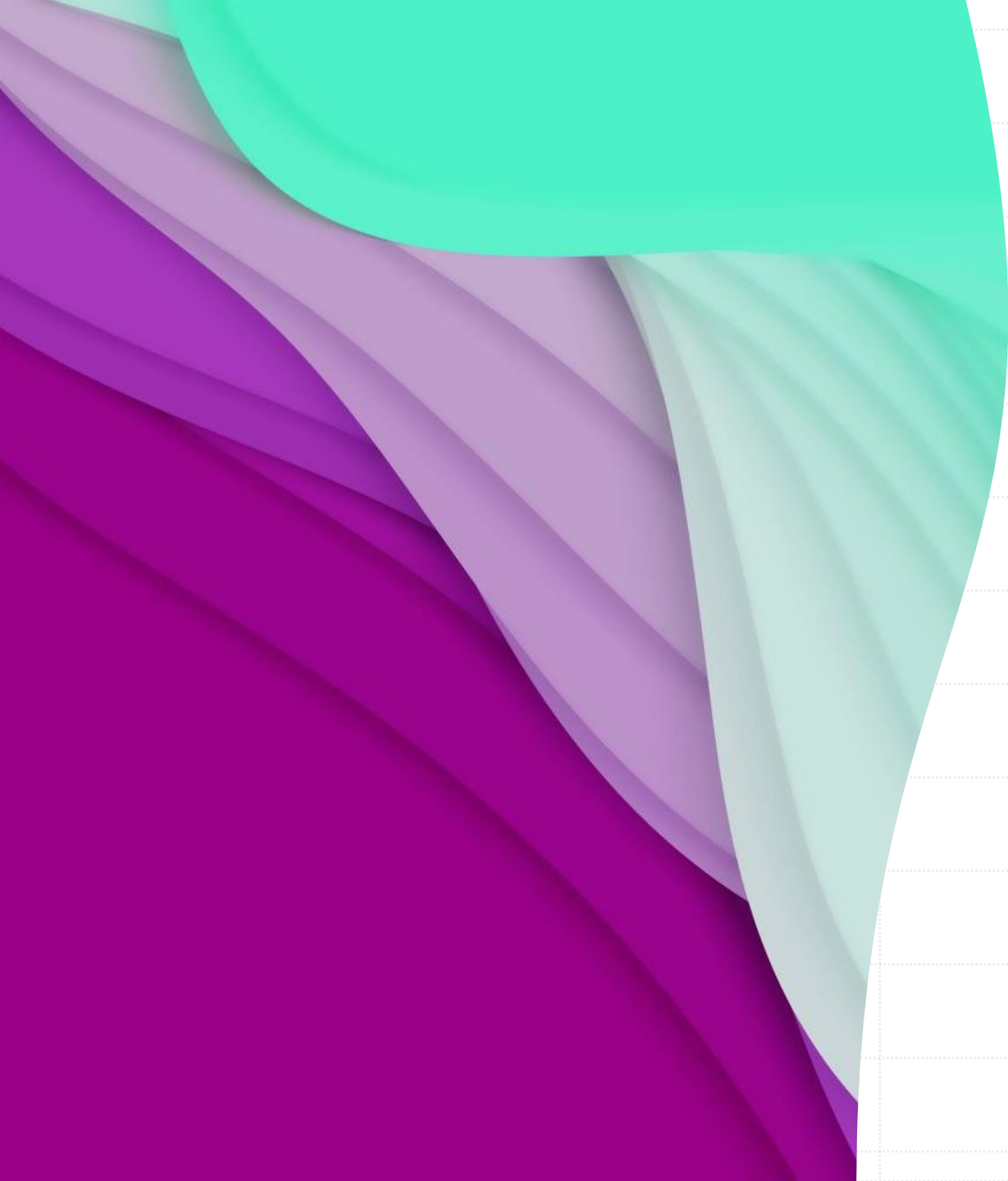






Data Visualization in Python

Week 07 - Day 02

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Objectives

- Understand **data visualization** basics.
- Explore Python **libraries** for visualization.
- Practice with **Matplotlib** and **Seaborn**.



Introduction to Data Visualization

- Definition and importance of data visualization in Python.
- **Key libraries:** Matplotlib, Seaborn, and Plotly



Installing Required Libraries

- Code: `pip install matplotlib seaborn plotly`
- Explanation of each library's primary use.



Getting Started with Matplotlib

- Overview of Matplotlib's capabilities.
- Code: Basic import
 - `import matplotlib.pyplot as plt`



Basic Plotting with Matplotlib

- Code example for a line plot:

```
import matplotlib.pyplot as plt  
plt.plot([1, 2, 3, 4], [10, 20, 25, 30])  
plt.show()
```

Customizing Plots - Titles and Labels

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4], [10, 20, 25, 30])
plt.title("Simple Line Plot")
plt.xlabel("Time")
plt.ylabel("Temperature")
plt.show()
```


Adding Legends and Grid

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4], [10, 20, 25, 30], label="Data Line")
plt.legend()
plt.grid(True)
plt.show()
```

Plot Types - Scatter Plot

```
import matplotlib.pyplot as plt
plt.scatter([1, 2, 3, 4], [30, 25, 20, 15])
plt.title("Scatter Plot Example")
plt.show()
```

Plot Types - Bar Plot

```
import matplotlib.pyplot as plt
plt.bar(['A', 'B', 'C'], [5, 10, 7])
plt.title("Bar Plot Example")
plt.show()
```

Working with Histograms

```
import matplotlib.pyplot as plt
data = [1, 2, 2, 3, 3, 3, 4, 5, 5]
plt.hist(data, bins=5)
plt.title("Histogram Example")
plt.show()
```

Pie Charts for Categorical Data

```
import matplotlib.pyplot as plt
labels = ['A', 'B', 'C']
sizes = [10, 20, 70]
plt.pie(sizes, labels=labels)
plt.show()
```



Introduction to Seaborn

- Overview of Seaborn and its connection to Matplotlib.
- Benefits for statistical data visualization

Seaborn - Loading Sample Data

```
import seaborn as sns  
data = sns.load_dataset("tips")  
print(data.head())
```

Seaborn - Scatter Plots with Regression Line

```
import matplotlib.pyplot as plt
import seaborn as sns
data = sns.load_dataset("tips")
sns.regplot(x="total_bill", y="tip", data=data)
plt.show()
```

Seaborn - Histogram and KDE

```
import matplotlib.pyplot as plt
import seaborn as sns
data = sns.load_dataset("tips")
sns.histplot(data['total_bill'], kde=True)
plt.show()
```

Seaborn - Pair Plot

```
import matplotlib.pyplot as plt
import seaborn as sns
data = sns.load_dataset("tips")
sns.pairplot(data)
plt.show()
```

Seaborn - Box Plot

```
import matplotlib.pyplot as plt
import seaborn as sns
data = sns.load_dataset("tips")
sns.boxplot(x="day", y="total_bill", data=data)
plt.show()
```



Introduction to Plotly

- Overview and benefits of interactive visualizations in Plotly.



Plotly – Simple Line Plot

Plotly - Interactive Bar Chart

```
import plotly.graph_objects as go
fig = go.Figure(go.Scatter(x=[1, 2, 3], y=[3, 1, 6]))
fig.write_html('first_figure.html', auto_open=True)
```


Plotly - Interactive Bar Chart

```
import plotly.graph_objects as go
fig = go.Figure(go.Bar(x=['A', 'B', 'C'], y=[20, 14, 23]))
fig.write_html('first_figure.html', auto_open=True)
```

Creating Subplots in Matplotlib

```
import matplotlib.pyplot as plt
fig, axs = plt.subplots(1, 2)
axs[0].plot([1, 2, 3], [1, 4, 9])
axs[1].plot([1, 2, 3], [9, 4, 1])
plt.show()
```

Annotations in Matplotlib

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3], [4, 5, 6])
plt.annotate("Peak", xy=(2, 5), xytext=(2, 5.5),
              arrowprops=dict(arrowstyle="->"))
plt.show()
```

Advanced Plotting in Seaborn - Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt
data = sns.load_dataset("tips")
numeric_data = data.select_dtypes(include='number')
correlation_matrix = numeric_data.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", linewidths=0.5)
plt.title("Correlation Heatmap of the 'tips' Dataset")
plt.show()
```

